

Technical Report: Finite-Time Centroid Tracking, Exponential Formation Stabilization, and Input-to-State Stability for Multi-Agent Swarms

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Abstract

This technical report provides a rigorous mathematical formulation of an extended swarm kinematic guidance model. The framework guarantees exact finite-time trajectory tracking of the swarm centroid along an arbitrary reference trajectory, independent of the formation geometry. By employing a hybrid inter-agent coupling term governed by true Euclidean inter-agent distance, the individual formation tracking errors are proved to converge globally and exponentially to zero; once the centroid has settled, the rate improves to $1 + \lambda_{\min}(M)/(D^\nu + \varepsilon)$, D being a bound on the swarm diameter, so that the coupling parameters enter the guarantee explicitly. Furthermore, an Input-to-State Stability (ISS) analysis is developed under bounded additive disturbances, establishing a clean, forward-invariant disturbance hyper-ball and an analytical bound on the finite entrance time. We show that none of the formation-channel results require an assumption on the centroid: the sliding-mode term is dissipative for the formation Lyapunov function, so every such bound — nominal and perturbed alike — holds from the initial time rather than only after centroid convergence. Finally, we establish that the same sliding-mode term renders the centroid channel *super-linearly* robust, confining the centroid error to a ball of radius $\bar{w}^{1/\beta}$ rather than the linear $\sqrt{n}\bar{w}$ that bounds the formation channel.

1 System Formulation and Variable Definitions

Consider a multi-agent system consisting of $n \geq 2$ mobile agents operating in a d -dimensional Euclidean space $\mathbb{R}^d$. Let the state of agent $i \in \{1, \dots, n\}$ at time $t \geq t_0$ be denoted by its physical position coordinate vector:

$$x_i(t) \in \mathbb{R}^d, \quad \forall i \in \{1, \dots, n\}. \quad (1)$$

1.1 Variable and Parameter Definitions

The foundational variables and parameters governing the multi-agent system are defined as follows:

1. **Swarm Centroid $c(t)$:** The geometric center of the swarm in $\mathbb{R}^d$ is defined by:

$$c(t) = \frac{1}{n} \sum_{i=1}^n x_i(t) \in \mathbb{R}^d. \quad (2)$$

2. **Reference Trajectory $r(t)$:** The desired global trajectory to be tracked by the swarm centroid is denoted by $r(t) \in \mathbb{R}^d$. We assume $r(t)$ is continuously differentiable with derivative $\dot{r}(t) \in \mathbb{R}^d$.

3. **Centroid Tracking Error** $\sigma(t)$: The deviation of the swarm centroid from the target reference trajectory is given by:

$$\sigma(t) = c(t) - r(t) \in \mathbb{R}^d. \quad (3)$$

4. **Time-Varying Formation Bias Vectors** $p_i(t)$: The assigned target spatial offset of agent i relative to the swarm centroid is denoted by $p_i(t) \in \mathbb{R}^d$, with continuous time derivative $\dot{p}_i(t) \in \mathbb{R}^d$. The formation biases are constrained to satisfy the **Centering Condition**:

$$\sum_{i=1}^n p_i(t) = \mathbf{0}_d, \quad \implies \quad \sum_{i=1}^n \dot{p}_i(t) = \mathbf{0}_d, \quad \forall t \geq t_0. \quad (4)$$

The formation is assumed to have bounded diameter,

$$D_p := \sup_{t \geq t_0} \max_{i \neq j} \|p_i(t) - p_j(t)\| < \infty. \quad (5)$$

5. **Individual Formation Tracking Error** $\delta_i(t)$: The tracking error of agent i relative to its assigned moving slot is defined by:

$$\delta_i(t) = x_i(t) - p_i(t) - r(t) \in \mathbb{R}^d. \quad (6)$$

The stacked collective formation error vector is denoted by $\boldsymbol{\delta}(t) = [\delta_1^T(t), \delta_2^T(t), \dots, \delta_n^T(t)]^T \in \mathbb{R}^{nd}$.

6. **Finite-Time Sliding-Mode Term** $\zeta(t)$: The non-smooth feedback signal $\zeta(t) \in \mathbb{R}^d$ acts on the centroid tracking error $\sigma(t)$. Its s -th component ($s \in \{1, \dots, d\}$) is defined by:

$$\zeta_s(t) = \text{sign}(\sigma_s(t)) |\sigma_s(t)|^\beta, \quad \forall s \in \{1, \dots, d\}, \quad (7)$$

where $\beta \in (0, 1)$ is a fractional exponent governing the finite-time convergence rate.

7. Interaction Coupling Parameters:

- $M \in \mathbb{R}^{d \times d}$: A constant, symmetric, positive-definite matrix ($M = M^T > 0$) with minimum eigenvalue $\lambda_{\min}(M) > 0$.
 - $\nu \in \mathbb{N}_+$: An integer exponent governing the inter-agent potential field decay rate.
 - $\varepsilon > 0$: A small positive regularizing constant preventing singular forces during agent collocation.
8. **External Disturbance** $w_i(t)$: An unknown, additive disturbance vector acting on agent i , assumed to be Lebesgue measurable and uniformly bounded by a known constant $\bar{w} \geq 0$:

$$\|w_i(t)\| \leq \bar{w}, \quad \forall i \in \{1, \dots, n\}, \quad \forall t \geq t_0. \quad (8)$$

1.2 The Hybrid Swarm Kinematic Model

The proposed hybrid kinematic model governing each agent $i \in \{1, \dots, n\}$ is formulated as:

$$\dot{x}_i(t) = -(x_i(t) - p_i(t) - r(t)) + \dot{r}(t) + \dot{p}_i(t) - \zeta(t) - \frac{M}{n} \sum_{j=1}^n \frac{(x_i(t) - p_i(t)) - (x_j(t) - p_j(t))}{\|x_i(t) - x_j(t)\|^\nu + \varepsilon} + w_i(t). \quad (9)$$

Using the tracking error definition (6), notice that:

$$(x_i(t) - p_i(t) - r_i(t)) - (x_j(t) - p_j(t) - r_j(t)) = \delta_i(t) - \delta_j(t), \quad (10)$$

$$x_i(t) - x_j(t) = (\delta_i(t) - \delta_j(t)) + (p_i(t) - p_j(t)). \quad (11)$$

Thus, the dynamics (9) in error coordinates become:

$$\dot{\delta}_i(t) = -\delta_i(t) - \zeta(t) - \frac{M}{n} \sum_{j=1}^n \frac{\delta_i(t) - \delta_j(t)}{\|x_i(t) - x_j(t)\|^\nu + \varepsilon} + w_i(t). \quad (12)$$

1.3 A Structural Identity

The following elementary identity is used repeatedly in the sequel. It is what allows the sliding-mode term ζ — which is common to all agents and therefore does not obviously interact with the *formation* error — to be handled exactly rather than assumed away.

Lemma 1 (Centroid–Formation Identity and Dissipativity of ζ). *Under the centering condition (4), the individual formation errors sum to the centroid error scaled by n :*

$$\sum_{i=1}^n \delta_i(t) = n \sigma(t), \quad \forall t \geq t_0. \quad (13)$$

Consequently, along any trajectory,

$$-\sum_{i=1}^n \delta_i^T(t) \zeta(t) = -n \sum_{s=1}^d |\sigma_s(t)|^{1+\beta} \leq 0. \quad (14)$$

Proof. Summing (6) over i and applying (4):

$$\sum_{i=1}^n \delta_i(t) = \sum_{i=1}^n x_i(t) - \sum_{i=1}^n p_i(t) - n r(t) = n c(t) - \mathbf{0}_d - n r(t) = n \sigma(t), \quad (15)$$

which is (13). Since $\zeta(t)$ carries no agent index, it factors out of the sum:

$$\sum_{i=1}^n \delta_i^T(t) \zeta(t) = \left(\sum_{i=1}^n \delta_i(t) \right)^T \zeta(t) = n \sigma^T(t) \zeta(t) = n \sum_{s=1}^d \sigma_s(t) \text{sign}(\sigma_s(t)) |\sigma_s(t)|^\beta. \quad (16)$$

Using $\sigma_s \text{sign}(\sigma_s) = |\sigma_s|$ gives $\sum_i \delta_i^T \zeta = n \sum_s |\sigma_s|^{1+\beta} \geq 0$, and negating yields (14). $\square$

2 Centroid Finite-Time Trajectory Tracking

Proposition 1 (Finite-Time Centroid Convergence). *Consider the nominal swarm kinematic system (9) with $w_i(t) \equiv \mathbf{0}_d$. Under the centering condition (4), the swarm centroid tracking error $\sigma(t) = c(t) - r(t)$ converges identically to zero in a bounded finite time $\tau_c > 0$, irrespective of the chosen formation offsets $p_i(t)$ and interaction matrix M .*

Proof. Differentiating the swarm centroid $c(t) = \frac{1}{n} \sum_{i=1}^n x_i(t)$ with respect to time yields:

$$\begin{aligned}\dot{c}(t) &= \frac{1}{n} \sum_{i=1}^n \dot{x}_i(t) \\ &= -\frac{1}{n} \sum_{i=1}^n (x_i(t) - p_i(t)) + r(t) + \dot{r}(t) + \frac{1}{n} \sum_{i=1}^n \dot{p}_i(t) - \zeta(t) \\ &\quad - \frac{M}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{(x_i(t) - p_i(t)) - (x_j(t) - p_j(t))}{\|x_i(t) - x_j(t)\|^\nu + \varepsilon}.\end{aligned}\tag{17}$$

By the centering condition (4), $\sum_{i=1}^n p_i(t) = \mathbf{0}_d$ and $\sum_{i=1}^n \dot{p}_i(t) = \mathbf{0}_d$. Consequently:

$$\frac{1}{n} \sum_{i=1}^n (x_i(t) - p_i(t)) = \frac{1}{n} \sum_{i=1}^n x_i(t) - \frac{1}{n} \sum_{i=1}^n p_i(t) = c(t).\tag{18}$$

Next, examine the double summation term in (17). Let $\Gamma_{ij} = \frac{\delta_i(t) - \delta_j(t)}{\|x_i(t) - x_j(t)\|^\nu + \varepsilon}$. Since $\|x_i - x_j\| = \|x_j - x_i\|$, the denominator is symmetric with respect to indices i and j , while the numerator is skew-symmetric ($\delta_i - \delta_j = -(\delta_j - \delta_i)$). Therefore, $\Gamma_{ij} = -\Gamma_{ji}$, which implies:

$$\sum_{i=1}^n \sum_{j=1}^n \Gamma_{ij} = \mathbf{0}_d.\tag{19}$$

Substituting these simplifications into (17) yields the decoupled centroid dynamics:

$$\dot{c}(t) = -c(t) + r(t) + \dot{r}(t) - \zeta(t).\tag{20}$$

Subtracting $\dot{r}(t)$ from both sides, the centroid error dynamics satisfy:

$$\dot{\sigma}(t) = -\sigma(t) - \zeta(t).\tag{21}$$

Because the system (21) is decoupled across coordinates, we consider the scalar Lyapunov function for the s -th component ($s \in \{1, \dots, d\}$):

$$V_s(\sigma_s) = \frac{1}{2} \sigma_s^2(t).\tag{22}$$

Taking the time derivative along the trajectories of (21):

$$\dot{V}_s = \sigma_s \dot{\sigma}_s = -\sigma_s^2 - \sigma_s \text{sign}(\sigma_s) |\sigma_s|^\beta = -2V_s - 2^{\frac{\beta+1}{2}} V_s^{\frac{\beta+1}{2}}.\tag{23}$$

Since $V_s \geq 0$, we establish the differential inequality:

$$\dot{V}_s \leq -2^{\frac{\beta+1}{2}} V_s^{\frac{\beta+1}{2}} = -\alpha V_s^\eta,\tag{24}$$

where $\eta = \frac{\beta+1}{2} \in (0.5, 1)$ and $\alpha = 2^\eta > 0$. Solving this differential inequality by separation of variables yields:

$$V_s^{1-\eta}(t) \leq V_s^{1-\eta}(t_0) - \alpha(1-\eta)(t-t_0).\tag{25}$$

Hence, $V_s(t)$ reaches zero identically at the finite time:

$$\tau_s = t_0 + \frac{V_s^{1-\eta}(t_0)}{\alpha(1-\eta)} = t_0 + \frac{|\sigma_s(t_0)|^{1-\beta}}{1-\beta}.\tag{26}$$

Defining the global finite settling time $\tau_c = \max_{s=1, \dots, d} \tau_s$, we have $\sigma(t) \equiv \mathbf{0}_d$ and $\zeta(t) \equiv \mathbf{0}_d$ for all $t \geq \tau_c$. $\square$

3 Global Exponential Formation Stabilization

We now analyze the nominal tracking error dynamics. By Lemma 1 no assumption on the centroid channel is needed, so the analysis runs from the initial time t_0 rather than from τ_c .

Proposition 2 (Exponential Formation Convergence). *Consider the nominal dynamics (9) with $w_i(t) \equiv \mathbf{0}_d$ under (4)–(5), and let τ_c be the centroid settling time (26). Then:*

(i) *For all $t \geq t_0$, independently of M , ν and ε ,*

$$\|\delta(t)\| \leq \|\delta(t_0)\| e^{-(t-t_0)}. \quad (27)$$

(ii) *For all $t \geq \tau_c$, with $D_0 := \sqrt{2}\|\delta(t_0)\| + D_p$ and*

$$\kappa_0 := \frac{\lambda_{\min}(M)}{D_0^\nu + \varepsilon} \geq 0, \quad (28)$$

the formation error converges at the faster, coupling-dependent rate

$$\|\delta(t)\| \leq \|\delta(\tau_c)\| e^{-(1+\kappa_0)(t-\tau_c)}. \quad (29)$$

Consequently, $\lim_{t \rightarrow \infty} x_i(t) - p_i(t) - r(t) = \mathbf{0}_d$ for all $i \in \{1, \dots, n\}$.

Proof. Setting $w_i \equiv \mathbf{0}_d$ in (12) and writing $a_{ij}(t) := (\|x_i(t) - x_j(t)\|^\nu + \varepsilon)^{-1} = a_{ji}(t) > 0$, the nominal error dynamics are

$$\dot{\delta}_i(t) = -\delta_i(t) - \zeta(t) - \frac{M}{n} \sum_{j=1}^n a_{ij}(t) (\delta_i(t) - \delta_j(t)). \quad (30)$$

Let $V_\delta := \frac{1}{2}\|\delta\|^2$ and $\Omega := \sum_{i=1}^n \sum_{j=1}^n a_{ij} \delta_i^T M (\delta_i - \delta_j)$. Along (30), using (14) for the ζ term,

$$\dot{V}_\delta = -\|\delta\|^2 - n \sum_{s=1}^d |\sigma_s|^{1+\beta} - \frac{\Omega}{n}. \quad (31)$$

Swapping the dummy indices in Ω and using $a_{ij} = a_{ji}$ gives $\Omega = -\sum_{i,j} a_{ij} \delta_j^T M (\delta_i - \delta_j)$; averaging the two representations,

$$\Omega = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n a_{ij} (\delta_i - \delta_j)^T M (\delta_i - \delta_j) \geq a_{\min}(t) \lambda_{\min}(M) \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \|\delta_i - \delta_j\|^2 \geq 0, \quad (32)$$

where $a_{\min}(t) := \min_{i \neq j} a_{ij}(t)$.

Part (i). The sliding-mode term is retained and no property of σ is assumed. The last two terms of (31) are non-positive, so $\dot{V}_\delta \leq -2V_\delta$ for all $t \geq t_0$; the comparison principle gives (27).

Part (ii). For $t \geq \tau_c$ we have $\sigma \equiv \mathbf{0}_d$ by Proposition 1, so the middle term of (31) vanishes and, by Lemma 1, $\sum_{i=1}^n \delta_i = \mathbf{0}_d$. For vectors with zero sum,

$$\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \|\delta_i - \delta_j\|^2 = n \sum_{i=1}^n \|\delta_i\|^2 - \left\| \sum_{i=1}^n \delta_i \right\|^2 = n \|\delta\|^2, \quad (33)$$

so (32) gives $\Omega/n \geq a_{\min}(t) \lambda_{\min}(M) \|\delta\|^2$ and

$$\dot{V}_\delta \leq -(1 + a_{\min}(t) \lambda_{\min}(M)) \|\delta\|^2, \quad t \geq \tau_c. \quad (34)$$

It remains to bound $a_{\min}(t)$ from below. For $i \neq j$, $x_i - x_j = (\delta_i - \delta_j) + (p_i - p_j)$ and $\|\delta_i - \delta_j\|^2 \leq 2\|\delta_i\|^2 + 2\|\delta_j\|^2 \leq 2\|\delta\|^2$, so by part (i) and (5)

$$\|x_i(t) - x_j(t)\| \leq \sqrt{2} \|\delta(t_0)\| + D_p = D_0, \quad \forall t \geq t_0. \quad (35)$$

Hence $a_{\min}(t) \geq (D_0^\nu + \varepsilon)^{-1}$ and $a_{\min}(t) \lambda_{\min}(M) \geq \kappa_0$. Substituting into (34) yields $\dot{V}_\delta \leq -2(1 + \kappa_0) V_\delta$ on $[\tau_c, \infty)$, and the comparison principle gives (29). $\square$

Remark 1 (Role of the coupling parameters). *Part (i) is attained already with $M = 0$ and holds from t_0 without waiting for the centroid; part (ii) is what the coupling buys once the centroid has settled. The guaranteed rate $1 + \lambda_{\min}(M)/(D_0^\nu + \varepsilon)$ makes the three parameters explicit: $\lambda_{\min}(M)$ enters linearly, so rotated or anisotropic M with the same smallest eigenvalue give the same guarantee; larger ν shrinks κ_0 when $D_0 > 1$, i.e. the guarantee degrades faster with swarm size; and ε matters only when it is comparable to D_0^ν . Since the argument may be restarted at any $t_1 \geq \tau_c$ with D_0 replaced by $\sqrt{2}\|\delta(t_1)\| + D_p$, the rate improves as the swarm contracts and approaches $1 + \lambda_{\min}(M)/(D_p^\nu + \varepsilon)$; this is the quantity to compare with a late-time slope fit of $\|\delta(t)\|$ in simulation.*

Corollary 1 (Deterministic Acceptance Time for Physical Docking). *Let $\varepsilon_{\text{tol}} > 0$ denote an arbitrary user-specified acceptance tolerance radius (e.g., physical docking socket clearance or payload gripper margin). All individual tracking errors satisfy $\|\delta_i(t)\| \leq \varepsilon_{\text{tol}}$ for all $t \geq T_{\text{dock}}$, where the deterministic finite docking time is explicitly given by:*

$$T_{\text{dock}} = t_0 + \max\left(0, \ln\left(\frac{\|\delta(t_0)\|}{\varepsilon_{\text{tol}}}\right)\right). \quad (36)$$

Proof. From Proposition 2, for any agent $i \in \{1, \dots, n\}$ and $t \geq t_0$:

$$\|\delta_i(t)\| \leq \|\delta(t)\| \leq \|\delta(t_0)\| e^{-(t-t_0)}. \quad (37)$$

To guarantee that $\|\delta_i(t)\| \leq \varepsilon_{\text{tol}}$, it suffices to enforce:

$$\|\delta(t_0)\| e^{-(t-t_0)} \leq \varepsilon_{\text{tol}} \implies e^{-(t-t_0)} \leq \frac{\varepsilon_{\text{tol}}}{\|\delta(t_0)\|}. \quad (38)$$

If $\|\delta(t_0)\| \leq \varepsilon_{\text{tol}}$, the condition is satisfied at $t = t_0$. Otherwise, taking the natural logarithm:

$$-(t - t_0) \leq \ln\left(\frac{\varepsilon_{\text{tol}}}{\|\delta(t_0)\|}\right) = -\ln\left(\frac{\|\delta(t_0)\|}{\varepsilon_{\text{tol}}}\right) \implies t \geq t_0 + \ln\left(\frac{\|\delta(t_0)\|}{\varepsilon_{\text{tol}}}\right). \quad (39)$$

Combining both cases proves the claim. Note that the docking guarantee is thus available immediately, without waiting for the centroid to settle. $\square$

4 Input-to-State Stability (ISS) Under Bounded Disturbances

In practical deployments, agents are subject to non-ideal conditions such as unmodeled dynamics, friction variations, and actuator noise. We now analyze the system under persistent, uniformly bounded disturbances $\|w_i(t)\| \leq \bar{w}$.

Proposition 3 (Robust Formation Tracking and ISS Invariant Set). *Consider the perturbed hybrid error dynamics (12) with $\|w_i(t)\| \leq \bar{w}$ for all $t \geq t_0$, under the centering condition (4). Then for all $t \geq t_0$ — with no assumption on the centroid channel — the collective tracking error $\boldsymbol{\delta}(t)$ is Input-to-State Stable (ISS) with respect to the disturbance input $w(t)$ and satisfies:*

$$\|\boldsymbol{\delta}(t)\| \leq \|\boldsymbol{\delta}(t_0)\|e^{-(t-t_0)} + \sqrt{n}\bar{w} \left(1 - e^{-(t-t_0)}\right), \quad \forall t \geq t_0. \quad (40)$$

Consequently, the system possesses the forward-invariant disturbance hyper-ball:

$$\mathcal{B}_{ISS} = \left\{ \boldsymbol{\delta} \in \mathbb{R}^{nd} \mid \|\boldsymbol{\delta}\| \leq \sqrt{n}\bar{w} \right\}, \quad (41)$$

which every trajectory enters in finite time and never leaves.

Proof. Let $t \geq t_0$. We retain the sliding-mode term $\zeta(t)$ throughout; no property of $\sigma(t)$ is invoked. Taking the time derivative of $V_\delta(\boldsymbol{\delta}) = \frac{1}{2} \sum_{i=1}^n \|\delta_i\|^2$ along the full perturbed dynamics (12):

$$\begin{aligned} \dot{V}_\delta &= \sum_{i=1}^n \delta_i^T(t) \left[-\delta_i(t) - \zeta(t) - \frac{M}{n} \sum_{j=1}^n \frac{\delta_i(t) - \delta_j(t)}{\|x_i(t) - x_j(t)\|^\nu + \varepsilon} + w_i(t) \right] \\ &= - \sum_{i=1}^n \|\delta_i(t)\|^2 - \underbrace{n \sum_{s=1}^d |\sigma_s(t)|^{1+\beta}}_{\leq 0 \text{ (Lemma 1)}} - \underbrace{\frac{1}{2n} \sum_{i=1}^n \sum_{j=1}^n \frac{(\delta_i - \delta_j)^T M (\delta_i - \delta_j)}{\|x_i - x_j\|^\nu + \varepsilon}}_{\leq 0 \text{ (since } M \succ 0)} + \sum_{i=1}^n \delta_i^T(t) w_i(t), \end{aligned} \quad (42)$$

where the ζ contribution has been evaluated using (14). Both bracketed terms are non-positive, so:

$$\dot{V}_\delta \leq -2V_\delta + \sum_{i=1}^n \|\delta_i(t)\| \|w_i(t)\| \leq -2V_\delta + \bar{w} \sum_{i=1}^n \|\delta_i(t)\|. \quad (43)$$

To relate the sum of norms $\sum_{i=1}^n \|\delta_i\|$ to the Lyapunov function V_δ , we apply the Cauchy–Schwarz inequality in $\mathbb{R}^n$ between the norm vector $u = [\|\delta_1\|, \dots, \|\delta_n\|]^T$ and the vector of ones $\mathbf{1}_n = [1, \dots, 1]^T$:

$$\sum_{i=1}^n \|\delta_i\| = \mathbf{1}_n^T u \leq \|\mathbf{1}_n\|_2 \|u\|_2 = \sqrt{n} \sqrt{\sum_{i=1}^n \|\delta_i\|^2} = \sqrt{n} \|\boldsymbol{\delta}\|. \quad (44)$$

Since $\|\boldsymbol{\delta}\| = \sqrt{2V_\delta}$, we obtain the strict inequality:

$$\sum_{i=1}^n \|\delta_i(t)\| \leq \sqrt{2nV_\delta(t)}. \quad (45)$$

Substituting this bound into (43) yields:

$$\dot{V}_\delta \leq -2V_\delta + \sqrt{2n}\bar{w}\sqrt{V_\delta}. \quad (46)$$

Step 1: Square-Root Transformation. For $V_\delta > 0$, define the auxiliary scalar variable $W(t) = \sqrt{V_\delta(t)} = \frac{1}{\sqrt{2}} \|\boldsymbol{\delta}(t)\|$. Differentiating $W(t)$ with respect to time:

$$\dot{W}(t) = \frac{\dot{V}_\delta(t)}{2\sqrt{V_\delta(t)}}. \quad (47)$$

Dividing (46) by $2\sqrt{V_\delta}$ gives:

$$\dot{W}(t) \leq \frac{-2V_\delta + \sqrt{2n\bar{w}}\sqrt{V_\delta}}{2\sqrt{V_\delta}} = -W(t) + \sqrt{\frac{n}{2}}\bar{w}. \quad (48)$$

Multiplying (48) by the integrating factor e^t and integrating from t_0 to t :

$$W(t) \leq W(t_0)e^{-(t-t_0)} + \sqrt{\frac{n}{2}}\bar{w} \left(1 - e^{-(t-t_0)}\right). \quad (49)$$

Multiplying through by $\sqrt{2}$ yields the state error norm bound (40):

$$\|\delta(t)\| \leq \|\delta(t_0)\|e^{-(t-t_0)} + \sqrt{n\bar{w}} \left(1 - e^{-(t-t_0)}\right). \quad (50)$$

Step 2: Invariance and Attractivity of $\mathcal{B}_{\text{ISS}}$. Rewriting the Lyapunov derivative bound (46) directly in terms of $\|\delta\|$:

$$\dot{V}_\delta \leq -\|\delta\| (\|\delta\| - \sqrt{n\bar{w}}). \quad (51)$$

We evaluate the vector field across three distinct state-space regions:

1. **Outside $\mathcal{B}_{\text{ISS}}$ ($\|\delta\| > \sqrt{n\bar{w}}$):** The term $(\|\delta\| - \sqrt{n\bar{w}}) > 0$, which ensures $\dot{V}_\delta < 0$ strictly. The stabilizing linear state feedback dominates the disturbance, forcing the error norm $\|\delta(t)\|$ to strictly decrease.
2. **On the Boundary $\partial\mathcal{B}_{\text{ISS}}$ ($\|\delta\| = \sqrt{n\bar{w}}$):** We have $\dot{V}_\delta \leq 0$. Because the gradient $\nabla V_\delta(\delta) = \delta$ points outward, $\dot{V}_\delta \leq 0$ proves that the vector field never points outside the boundary surface. Hence, $\mathcal{B}_{\text{ISS}}$ is a forward-invariant set.
3. **Inside $\mathcal{B}_{\text{ISS}}$ ($\|\delta\| < \sqrt{n\bar{w}}$):** The sign of $\dot{V}_\delta$ is indefinite. Persistent disturbances can inject energy when tracking errors are small ($\delta \approx \mathbf{0}_{nd}$), causing bounded chattering. However, because $\dot{V}_\delta \leq 0$ on $\partial\mathcal{B}_{\text{ISS}}$, trajectories cannot escape $\mathcal{B}_{\text{ISS}}$.

Taking the limit as $t \rightarrow \infty$ in (40) establishes ultimate boundedness:

$$\limsup_{t \rightarrow \infty} \|\delta(t)\| \leq \sqrt{n\bar{w}}. \quad (52)$$

□

Corollary 2 (Finite Entrance Time into the ε_{tol} -Neighborhood of $\mathcal{B}_{\text{ISS}}$). *For any positive clearance margin $\varepsilon_{\text{tol}} > 0$, the collective tracking error $\delta(t)$ enters the neighborhood $\mathcal{B}_\varepsilon = \{\delta \mid \|\delta\| \leq \sqrt{n\bar{w}} + \varepsilon_{\text{tol}}\}$ at or before the finite timestamp:*

$$T_{\text{enter}} = t_0 + \max \left(0, \ln \left(\frac{\max(0, \|\delta(t_0)\| - \sqrt{n\bar{w}})}{\varepsilon_{\text{tol}}} \right) \right). \quad (53)$$

Proof. From Proposition 3, for all $t \geq t_0$:

$$\|\delta(t)\| \leq \sqrt{n\bar{w}} + (\|\delta(t_0)\| - \sqrt{n\bar{w}}) e^{-(t-t_0)}. \quad (54)$$

If $\|\delta(t_0)\| \leq \sqrt{n\bar{w}} + \varepsilon_{\text{tol}}$, the state is already within $\mathcal{B}_\varepsilon$ at $t = t_0$. If $\|\delta(t_0)\| > \sqrt{n\bar{w}} + \varepsilon_{\text{tol}}$, setting the upper bound less than or equal to $\sqrt{n\bar{w}} + \varepsilon_{\text{tol}}$ gives:

$$(\|\delta(t_0)\| - \sqrt{n\bar{w}}) e^{-(t-t_0)} \leq \varepsilon_{\text{tol}} \implies t \geq t_0 + \ln \left(\frac{\|\delta(t_0)\| - \sqrt{n\bar{w}}}{\varepsilon_{\text{tol}}} \right), \quad (55)$$

which completes the proof. □

4.1 Centroid Robustness: Super-Linear Attenuation of the Sliding Term

Proposition 3 bounds the *formation* channel, and does so linearly in $\bar{w}$. The *centroid* channel — which carries the mission-level tracking objective — is protected considerably more strongly, because ζ acts on it directly rather than through the collective error.

Proposition 4 (Centroid Disturbance Ball). *Consider (9) with $\|w_i(t)\| \leq \bar{w}$, $\bar{w} > 0$, under the centering condition (4). Let $\bar{w}_c(t) = \frac{1}{n} \sum_{i=1}^n w_i(t)$ denote the mean disturbance, so that $\|\bar{w}_c(t)\| \leq \bar{w}$ and hence $|\bar{w}_{c,s}(t)| \leq \bar{w}$ for each s . Then:*

(i) *The centroid error obeys the perturbed, coordinate-decoupled dynamics*

$$\dot{\sigma}_s(t) = -\sigma_s(t) - \text{sign}(\sigma_s(t))|\sigma_s(t)|^\beta + \bar{w}_{c,s}(t), \quad s \in \{1, \dots, d\}. \quad (56)$$

(ii) *The set $\mathcal{S} = \{\sigma \in \mathbb{R}^d \mid |\sigma_s| \leq \bar{w}^{1/\beta}, s = 1, \dots, d\}$ is forward invariant.*

(iii) *For every $\mu \in (0, 1)$, $\sigma(t)$ enters the set $\mathcal{S}_\mu = \{\sigma \mid |\sigma_s| \leq (\bar{w}/\mu)^{1/\beta}\}$ no later than the finite time*

$$T_\sigma(\mu) = t_0 + \max_{s=1, \dots, d} \frac{|\sigma_s(t_0)|^{1-\beta}}{(1-\mu)(1-\beta)}. \quad (57)$$

(iv) *Consequently $\limsup_{t \rightarrow \infty} |\sigma_s(t)| \leq \bar{w}^{1/\beta}$ for every s , and $\limsup_{t \rightarrow \infty} \|\sigma(t)\| \leq \sqrt{d} \bar{w}^{1/\beta}$.*

Proof. (i) Averaging (9) over i and repeating the two simplifications of Proposition 1 — the centering condition (4) for the bias terms, and the skew-symmetry $\Gamma_{ij} = -\Gamma_{ji}$ for the double sum — leaves the disturbance mean as the only additional contribution:

$$\dot{c}(t) = -c(t) + r(t) + \dot{r}(t) - \zeta(t) + \bar{w}_c(t), \quad (58)$$

and subtracting $\dot{r}(t)$ gives $\dot{\sigma} = -\sigma - \zeta + \bar{w}_c$, which is (56) componentwise.

(ii) Take $V_s = \frac{1}{2}\sigma_s^2$. Along (56),

$$\dot{V}_s = -\sigma_s^2 - |\sigma_s|^{1+\beta} + \sigma_s \bar{w}_{c,s} \leq -\sigma_s^2 - |\sigma_s| \left(|\sigma_s|^\beta - \bar{w} \right). \quad (59)$$

On the boundary $|\sigma_s| = \bar{w}^{1/\beta}$ we have $|\sigma_s|^\beta = \bar{w}$, so the second term vanishes and

$$\dot{V}_s \leq -\sigma_s^2 = -\bar{w}^{2/\beta} < 0. \quad (60)$$

The vector field therefore points strictly inward on $\partial\mathcal{S}$, and since (56) is decoupled across s , the box $\mathcal{S}$ is forward invariant.

(iii) Suppose $|\sigma_s| \geq (\bar{w}/\mu)^{1/\beta}$, equivalently $\bar{w} \leq \mu|\sigma_s|^\beta$. Substituting into (59):

$$\dot{V}_s \leq -\sigma_s^2 - |\sigma_s|^{1+\beta} + \mu|\sigma_s|^{1+\beta} = -2V_s - (1-\mu)2^{\frac{\beta+1}{2}}V_s^{\frac{\beta+1}{2}} \leq -(1-\mu)\alpha V_s^\eta, \quad (61)$$

with $\eta = \frac{\beta+1}{2}$ and $\alpha = 2^\eta$ exactly as in Proposition 1. This is the differential inequality of Proposition 1 with α replaced by $(1-\mu)\alpha$; repeating the separation-of-variables step yields (57). Outside $\mathcal{S}_\mu$ the decay is therefore of finite-time type, so $\mathcal{S}_\mu$ is reached in finite time and, by the same argument as in (ii) applied at radius $(\bar{w}/\mu)^{1/\beta}$, not left thereafter.

(iv) Letting $\mu \uparrow 1$ in (iii) and combining with the forward invariance of $\mathcal{S}$ from (ii) gives the component-wise limit; the norm bound follows from $\|\sigma\| \leq \sqrt{d} \max_s |\sigma_s|$. $\square$

Table 1: Summary of Swarm Mathematical Properties under the Hybrid Kinematic Model

Property	State Variable	Convergence Type	Analytical Bound
Centroid Tracking	$\sigma(t) = c(t) - r(t)$	Finite-Time (τ_c)	$\tau_c = \max_s \left(t_0 + \frac{ \sigma_s(t_0) ^{1-\beta}}{1-\beta} \right)$
Nominal Formation	$\delta(t) \ (w_i \equiv \mathbf{0})$	Exponential ($\rightarrow \mathbf{0}$)	$\ \delta(t)\ \leq \ \delta(t_0)\ e^{-(t-t_0)}$
Nominal Formation, $t \geq \tau_c$	$\delta(t) \ (w_i \equiv \mathbf{0})$	Exponential, rate $1 + \kappa_0$	$\kappa_0 = \lambda_{\min}(M)/(D_0^\nu + \varepsilon)$, Prop. 2(ii)
Physical Docking	$\ \delta_i(t)\ \leq \varepsilon_{\text{tol}}$	Deterministic Time (T_{dock})	$T_{\text{dock}} = t_0 + \ln(\ \delta(t_0)\ /\varepsilon_{\text{tol}})$
Perturbed Formation	$\delta(t) \ (\ w_i\ \leq \bar{w})$	ISS / Invariant Hyper-ball	$\mathcal{B}_{\text{ISS}} = \{\delta \mid \ \delta\ \leq \sqrt{n}\bar{w}\}$
Perturbed Centroid	$\sigma(t) \ (\ w_i\ \leq \bar{w})$	Invariant Ball (super-linear)	$\mathcal{S} = \{\sigma \mid \sigma_s \leq \bar{w}^{1/\beta}\}$

5 Summary of Theoretical Properties

Expanding $\|\delta_i - \delta_j\|^2 = \|\delta_i\|^2 + \|\delta_j\|^2 - 2\delta_i^T \delta_j$ and summing over all i, j ,

$$\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \|\delta_i - \delta_j\|^2 = n \sum_{i=1}^n \|\delta_i\|^2 - \sum_{i=1}^n \sum_{j=1}^n \delta_i^T \delta_j = n \|\delta\|^2 - \left\| \sum_{i=1}^n \delta_i \right\|^2 = n \|\delta\|^2, \quad (62)$$

where the last step uses $\sum_i \delta_i = \mathbf{0}_d$.